

Ferroptosis Signature Correlates with Ovarian Cancer Prognosis and Chemotherapy Response

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Background: Ovarian cancer is a leading cause of gynecological cancer mortality, with late diagnosis, high recurrence and chemotherapy resistance closely linked to the tumor microenvironment (TME). Ferroptosis, an iron-dependent regulated cell death, is a promising therapeutic target, but its role in ovarian cancer TME remodeling and treatment resistance remains unclear.

Methods: We integrated bulk transcriptome and single-cell multi-omics datasets from ovarian cancer patient cohorts. Weighted gene co-expression network analysis (WGCNA) and machine learning algorithms were applied to develop and externally validate a ferroptosis-related risk signature (FRS) across independent cohorts including TCGA-OV and GSE14764. We systematically analyzed correlations between FRS and clinical outcomes, TME immune landscape, somatic genomic aberrations, as well as computationally predicted chemotherapeutic susceptibility. Subsequent in vitro assays using two ovarian cancer cell lines (SKOV3 and Caov-3) were conducted to preliminarily explore the combined anti-tumor activity of Erastin and paclitaxel, alongside underlying molecular associations.

Results: We established a 17-gene-based Ferroptosis Sensitivity Score (FRS), with a mean index of 0.741 across the cohort. FRS effectively stratified patients into high/low-risk groups with significant survival differences; high-risk patients had a suppressive immune TME, enriched tumor-promoting pathways and distinct genomic alterations. FRS was an independent prognostic biomarker, and a nomogram integrating FRS and clinical features improved survival prediction. In vitro, Erastin dose-dependently upregulated ferroptosis-associated proteins in SKOV3 and Caov-3 cells; combining Erastin with paclitaxel alleviated paclitaxel resistance, induced apoptosis, and transcriptomic analysis showed DEGs enriched in cell division, cell cycle, MAPK and lipid metabolism pathways, confirming their synergistic anti-tumor effect via regulating ferroptosis, apoptosis and multiple signaling pathways.

Conclusion: This work constructs a ferroptosis-derived risk signature with prognostic and chemoresponse predictive value for ovarian cancer, supported by multi-omics cohort analysis. Preliminary in vitro data from two ovarian cancer cell lines (SKOV3 and Caov-3) imply combinatorial Erastin-paclitaxel treatment may exert synergistic anti-tumor effects and alleviate paclitaxel resistance. Our findings offer preliminary clues correlating ferroptosis with TME remodeling and lay a preliminary theoretical foundation for exploring combinatorial regimens to counter chemoresistance in ovarian cancer.

Keywords: ferroptosis, ovarian cancer, prognostic signature, tumor microenvironment

Introduction

Ovarian cancer ranks among the most lethal malignancies affecting women worldwide. It encompasses several histological subtypes, among which epithelial ovarian cancer (EOC) is the most prevalent. The majority of patients are diagnosed at an advanced stage, at which point treatment options are limited and recurrence rates are high.¹ Despite advances in therapy, reliable biomarkers for early detection and prediction of therapy resistance remain elusive. Therefore, more effective diagnostic methods and treatment strategies are urgently needed to improve patient survival outcomes.

The standard management of epithelial ovarian cancer (EOC) typically consists of aggressive cytoreductive surgery followed by platinum-based chemotherapy, most frequently using carboplatin and paclitaxel. However, the majority of

patients experience disease recurrence and ultimately acquire platinum resistance.² This pattern of recurrence remains a major obstacle to long-term disease control, underscoring the need for alternative therapeutic strategies.

In recent years, ferroptosis has gained significant attention in cancer research.^{3,4} This iron-dependent form of regulated cell death is driven by lipid peroxidation and is mechanistically distinct from apoptosis or necrosis. Key regulators—including system Xc⁻/SLC7A11, GPX4, FSP1, members of the ACSL family, and iron-handling proteins—orchestrate the maintenance of redox balance and lipid metabolism, thereby determining cellular susceptibility to ferroptosis.⁵ Notably, ovarian cancer harbors unique disease-specific biological features that render ferroptosis particularly meaningful for this malignancy, including intrinsic redox adaptation, aberrant lipid metabolism, and therapeutic vulnerabilities linked to BRCA/HRD deficiency, as well as distinct response patterns to platinum and paclitaxel regimens that distinguish it from other solid tumor types.^{6,7}

Accumulating experimental evidence indicates that modulating ferroptosis pathways can significantly enhance therapeutic efficacy. For instance, inhibition of GPX4 sensitizes BRCA1-deficient ovarian cancer models to PARP inhibitors.⁸ Similarly, small molecules such as artesunate derivatives can trigger ferroptosis via transcriptional cascades involving HOXC11-PROM2-PI3K/AKT, leading to suppression of tumor growth.⁹ Collectively, these findings suggest that ferroptosis activation represents a promising strategy to overcome therapy resistance, though clinical validation is still limited.

The tumor microenvironment (TME) critically regulates ferroptosis and influences treatment response. Ferroptotic cell death can remodel immune infiltration and enhance tumor antigenicity, while macrophages, cytokines, and local iron metabolism reciprocally modulate cellular susceptibility to ferroptosis.¹⁰ A comprehensive understanding of this bidirectional crosstalk is crucial for harnessing ferroptosis in effective therapeutic strategies.

To elucidate these complex mechanisms, integrated multi-omics analysis combined with machine learning (ML) has emerged as a powerful strategy. The integration of transcriptomic, proteomic, and single-cell data enables precise characterization of molecular networks associated with ferroptosis. Leveraging these data, ML techniques can extract predictive features, derive robust ferroptosis-related signatures (FRS), and construct prognostic models with cross-cohort generalizability.¹¹ Recent studies applying multi-omics and ML in ovarian and other cancers have successfully identified programmed cell death signatures and immunometabolic correlates predictive of prognosis and treatment response, underscoring the feasibility and translational value of this approach.¹² Nevertheless, existing ferroptosis-based prognostic signatures specific to ovarian cancer have prominent limitations: most models rely merely on bulk transcriptome data without single-cell TME validation, rarely link risk stratification to platinum and paclitaxel chemosensitivity, and lack supporting in vitro functional verification of ferroptosis-targeted combinatorial therapy. This unmet research gap restricts our full understanding of ferroptosis-mediated TME remodeling and chemoresistance in ovarian cancer.

Against this research backdrop, the present study consists of three distinct components that clearly separate bioinformatic analyses from wet-lab experiments: we first establish and validate an ovarian cancer prognostic signature linked to ferroptosis, then systematically characterize tumor microenvironment immune patterns associated with ferroptosis risk, and finally conduct preliminary in vitro functional assessments of Erastin combined with paclitaxel in SKOV3 and Caov-3 cells. We selected SKOV3 and Caov-3 ovarian cancer cell lines for in vitro functional assays for clear biological rationale: SKOV3 cells are well-established paclitaxel-resistant ovarian cancer models, whereas Caov-3 cells represent chemotherapy-sensitive epithelial ovarian cancer cells, enabling us to evaluate the synergistic anti-resistance effect of Erastin plus paclitaxel across distinct chemosensitivity backgrounds.

Considering all above analyses, this study provides a reliable prognostic biomarker derived from multi-omics cohort data and preliminary in vitro experimental clues for ferroptosis-based combinatorial regimens, offering mechanistic insights into ferroptosis-driven TME remodeling and tentative preclinical directions to tackle ovarian cancer chemotherapy resistance.

Materials and Methods

Data Collection and Processing

RNA sequencing data and corresponding clinical information for ovarian cancer were obtained from The Cancer Genome Atlas (TCGA) and the Genotype-Tissue Expression (GTEx) project. Gene expression levels were quantified using

fragments per kilobase of transcript per million mapped reads (FPKM) values.¹³ Samples with incomplete clinicopathological data were excluded, and genes with an average expression below 0.1 were removed. To eliminate cross-platform batch bias between TCGA tumor and GTEx normal tissues, we implemented ComBat batch correction, followed by log2 transformation of normalized FPKM expression values for uniform downstream analysis. After filtering, 372 TCGA-OV patient samples were retained for subsequent analysis. Additionally, single-cell RNA sequencing datasets (GSE154600, GSE158937, and GSE181955) were downloaded from the Gene Expression Omnibus (GEO) database. To assess the prognostic significance of the identified markers, GEO datasets, GSE14764 (80 samples), were included. Additionally, a curated compilation of 1,393 genes associated with ferroptosis was derived from existing literature.¹⁴ All publicly available datasets in this section are explicitly listed with unique accession numbers in the Data Availability statement of the manuscript for transparency and reproducibility.

Single-Cell RNA Sequencing Data Processing

Single-cell RNA sequencing data from seven patients with early-stage ovarian cancer, including datasets GSE154600, GSE158937, and GSE181955, were analyzed using the Seurat software package.¹³ Quality control was performed by excluding cells with mitochondrial gene content exceeding 10% and retaining those with total gene counts between 200 and 6,000. After strict QC filtering, a total of 42,687 valid single cells were reserved for subsequent analysis. To identify highly variable genes (HVGs), we ran the FindVariableFeatures function with the variance-stabilizing transformation (VST) algorithm embedded in Seurat. Genes were sorted by their standardized dispersion values, and the top 2000 genes with the highest dispersion scores were retained as variable genes for dimensionality reduction and downstream exploration. Batch effects across samples were corrected using the Harmony package.¹⁵ Cell clustering was performed with the “FindNeighbors” and “FindClusters” functions, and results were visualized via UMAP. Cell types were annotated by integrating canonical ovarian tissue cell marker panels, published single-cell ovarian cancer reference profiles.¹⁶ Gene set activity was evaluated in individual cells using the “AddModuleScore” function, with statistical significance of differentially expressed genes (DEGs) assessed by the Wilcoxon test.¹⁷ Furthermore, ferroptosis-related DEGs identified from single-cell data were incorporated into weighted gene co-expression network analysis (WGCNA) of bulk transcriptomic data.^{18,19}

Enrichment Analysis

Single-sample gene set enrichment analysis (ssGSEA) was employed to quantify the enrichment level of specific gene sets in individual samples, reflecting the extent of their upregulation or downregulation. In this study, ferroptosis activity scores were calculated for each sample in the TCGA-OV cohort using the “GSVA” R package. To investigate pathways associated with the identified features, we utilized the “limma” package to identify pathways showing significant differences between high- and low-risk groups, and computed GSVA scores for 50 critical pathways.²⁰ We used h.all.v2023.1.Hs.symbols.gmt for Hallmark gene set enrichment and performed separate GO term enrichment via clusterProfiler with the official GO annotation. Independent FDR corrections were applied for these two enrichment analyses to prevent potential confounding. Significance thresholds set at $FDR < 0.25$ and $|NES| > 1$. These analyses were conducted using the “clusterProfiler” R package.^{21,22}

Weighted Gene Co-Expression Network Analysis (WGCNA)

Weighted Gene Co-expression Network Analysis (WGCNA) was employed to identify gene correlation patterns across samples and to detect modules of highly co-expressed genes.²³ We explicitly selected a soft-thresholding power $\beta = 6$ to satisfy scale-free network topology criteria ($R^2 > 0.85$). A dynamic tree-cutting algorithm was then applied to cluster genes and define co-expression modules. Finally, the modules most strongly correlated with the ferroptosis score were identified for subsequent investigation.

Construction of Prognostic Signature Using Machine Learning Techniques

Differential expression analysis was performed on the TCGA-OV RNA-seq dataset using the “limma” package, with thresholds set at $|\log FC| > 0.5$ and adjusted p -value < 0.05 .²⁴ The obtained differentially expressed genes (DEGs) were

intersected with ferroptosis-associated hub genes screened from WGCNA, and the overlapping gene set was retained as candidate genes for prognostic model construction. We adopted multiple mainstream machine learning algorithms, including LASSO Cox regression, random forest, and support vector machine (SVM), to screen candidate prognostic genes and build the ferroptosis-related signature. For hyperparameter tuning of each algorithm, we implemented 5-fold cross-validation repeated 10 times on the training cohort to avoid overfitting and optimize predictive performance: For LASSO Cox regression: The penalty coefficient λ was tuned via cross-validation, and the λ value corresponding to the minimum partial likelihood deviance was selected as the optimal hyperparameter to compress redundant gene coefficients toward zero. For random forest: Grid search was applied to tune core hyperparameters including the number of decision trees (ntree, range: 50–500) and the number of candidate variables per split (mtry, range: 1–20), with the combination yielding the highest C-index chosen. For SVM with radial basis function kernel: Grid search was used to optimize the penalty parameter C (range: 10^{-3} – 10^3) and kernel width gamma (range: 10^{-4} – 10^2), selecting the hyperparameter pair that minimized cross-validation prediction error. After hyperparameter optimization, genes consistently identified as predictive by all three algorithms were retained to form the final ferroptosis-related risk signature (FRS). This pipeline generated a streamlined panel of ferroptosis-relevant genes with robust prognostic capacity for subsequent survival stratification and multi-dimensional validation.

Ferroptosis-Related Signatures

To construct a robust prognostic model, univariate Cox regression was first performed in the GSE14764 cohort to identify ferroptosis-related genes with prognostic relevance, which were subsequently validated in the TCGA cohort. Ten machine learning algorithms were combined into 101 distinct model configurations for variable selection and model building, including ridge regression, Lasso regression, stepwise Cox regression, CoxBoost, random survival forest (RSF), elastic net (Enet), plsRcox, SuperPC, survival SVM, and GBM. The predictive performance of each model was evaluated on both training and validation sets using the concordance index (C-index), which ranges from 0 to 1, with values closer to 1 indicating better predictive accuracy.^{25,26} All models were ranked according to their mean C-index to compare predictive efficacy. Algorithm combinations exhibiting strong predictive power and clinical relevance were retained for further analysis.^{27,28} The module most strongly associated with the ferroptosis score was subsequently identified for downstream investigation.

Survival Analysis and Development of a Predictive Nomogram

Kaplan-Meier survival analysis was performed using the “survminer” R package to assess differences in overall survival (OS), progression-free survival (PFS), and disease-specific survival (DSS) between high- and low-risk groups, with significance evaluated by the Log rank test ($P < 0.05$).²⁹ ROC curve analysis was conducted using the “timeROC” package to compare the area under the curve (AUC) of the FRS against other clinical parameters. To further elucidate the clinical relevance of the FRS, we analyzed its correlations with multiple clinicopathological features and performed Cox regression analyses in both the GSE14764 and TCGA-OV cohorts. To improve prognostic precision, a nomogram was constructed by integrating the FRS, clinical characteristics, and survival data to predict individualized survival outcomes in ovarian cancer patients. The discriminative ability and predictive accuracy of the nomogram were assessed using ROC curves, the concordance index (C-index), and calibration plots, while decision curve analysis (DCA) was applied to evaluate its net clinical benefit.³⁰

Examination of Genomic Variations Among FRS Risk Subgroups

Mutant Allele Tumor Heterogeneity (MATH) scores were calculated to assess intratumoral heterogeneity,³¹ whose prognostic value has been established across multiple tumor types.³² In this study, we computed MATH scores for each ovarian cancer patient and performed corresponding survival analyses. The R package “maftools” was employed to generate waterfall plots depicting the mutational profiles of both high-risk and low-risk cohorts.³³ Additionally, we identified 15 genes showing the most significant differences in copy number variation (CNV) between the risk subgroups.

Significance of FRS in Predicting Drug Sensitivity

The “pRRophetic” R package was employed to predict chemosensitivity in ovarian cancer patients stratified by FRS risk scores.³⁴ Drug response was assessed by half-maximal inhibitory concentration (IC₅₀) values, with differences attaining a *P*-value < 0.05 considered statistically significant.³⁵ We clarified that pRRophetic-derived IC₅₀ predictions serve only as preliminary computational hypothesis screening, and all drug sensitivity conclusions require subsequent cellular experimental verification rather than direct clinical inference.

Cell Culture and Drug Treatment

Human ovarian cancer cell lines SKOV3 and Caov-3 were both purchased from Shanghai Fuhheng Biotechnology Co., Ltd. The cells were cultured in Dulbecco’s Modified Eagle Medium (DMEM, Gibco, USA) supplemented with 10% fetal bovine serum (FBS, Thermo Fisher Scientific, USA) and 1% penicillin-streptomycin (Thermo Fisher Scientific, USA), and maintained at 37 °C in a humidified incubator with 5% CO₂. Ferroptosis inducer Erastin and paclitaxel were purchased from MedChemExpress (MCE, Monmouth Junction, NJ, USA). Both drugs were dissolved in dimethyl sulfoxide (DMSO, Sigma-Aldrich, USA) to prepare stock solutions, and then diluted to the required working concentrations with complete culture medium before cell treatment, with the final concentration of DMSO in the medium ≤ 0.1% (a non-toxic concentration to SKOV3/Caov-3 cells). The rationale for selecting 4 μM Erastin and 10 μM paclitaxel for combination treatment was determined via Western blot detection of core ferroptosis markers under an Erastin concentration gradient (0 μM, 2 μM, 4 μM, 8 μM) in both SKOV3 and Caov-3 ovarian cancer cell lines. For gradient concentration treatment, SKOV3 cells were exposed to serial concentrations of Erastin for 48 h to detect the expression of ferroptosis-associated proteins. For combination treatment, cells were co-incubated with 4 μM Erastin plus 10 μM paclitaxel for 48 h; in addition, SKOV3 cells were treated with various concentrations of paclitaxel alone or in combination with Erastin for subsequent cell proliferation assays. Bulk RNA-seq included only three biological replicates per group, restricting statistical power to identify DEGs with mild fold changes.

SiRNA Transfection

Specific knockdown of *BTN3A2* was achieved using human *BTN3A2* siRNA (sc-95160, Santa Cruz Biotechnology), with non-targeting control siRNA (sc-37007) served as the negative control. All transfection procedures were performed following the manufacturer’s standard protocols using the matched Santa Cruz transfection system, including siRNA Transfection Reagent (sc-29528), siRNA Transfection Medium (sc-36868), and siRNA Dilution Buffer (sc-29527) to ensure optimal transfection efficiency. Briefly, ovarian cancer cells were seeded 24 h before transfection and grown to 50%–60% confluence at the time of transfection. Prior to transfection, the complete culture medium was removed and cells were gently washed with sterile PBS. *BTN3A2* siRNA and control siRNA were separately diluted in siRNA Dilution Buffer, followed by gentle mixing with siRNA Transfection Reagent and incubation at room temperature to form stable transfection complexes. The prepared complexes were diluted in siRNA Transfection Medium and evenly added to the cell culture plates. After incubation at 37 °C with 5% CO₂ for 6 h, the transfection medium was replaced with fresh complete medium. Cells were continuously cultured for 48 h and then harvested for Western blotting to verify the knockdown efficiency of *BTN3A2*.

Western Blot Analysis

Total cellular proteins were extracted from treated SKOV3 cells using RIPA lysis buffer (Beyotime, China) containing protease and phosphatase inhibitors (Roche, Switzerland). The protein concentration was quantified with a BCA Protein Assay Kit (Thermo Fisher Scientific, USA). Equal amounts of total protein (30 μg per lane) were separated by 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and then transferred onto polyvinylidene fluoride (PVDF) membranes (Millipore, USA). The membranes were blocked with 5% non-fat milk in Tris-buffered saline with Tween 20 (TBST) for 1 h at room temperature, followed by incubation with primary antibodies against ferroptosis-associated proteins (SLC7A11, GPX4, FTH1), apoptosis-related proteins (Bax, BCL2) and internal reference GAPDH (all from Cell Signaling Technology, USA) at 4 °C overnight. After three washes with TBST, the membranes were incubated with horseradish peroxidase (HRP)-conjugated secondary antibodies (Hangzhou Huaan Biotechnology

Co., Ltd., China) for 1 h at room temperature. The protein bands were visualized using an Enhanced Chemiluminescence (ECL) Detection Kit (Glpbio, USA), and the relative expression levels of target proteins were normalized to GAPDH.

MTT Cell Proliferation Assay

The MTT assay was used to evaluate the effect of different drug treatments on SKOV3 cell proliferation. Briefly, SKOV3 cells were seeded into 96-well plates at a density of 5×10^3 cells per well and cultured overnight for cell adhesion. Then the cells were treated with various concentrations of paclitaxel alone or in combination with Erastin for 72 h. After treatment, 20 μ L of MTT solution (5 mg/mL, Sigma-Aldrich, USA) was added to each well, and the plates were incubated for an additional 4 h at 37 °C. The supernatant was carefully discarded, and 150 μ L of DMSO was added to each well to dissolve the formazan crystals formed by viable cells. The absorbance at 570 nm was measured using a microplate reader (Bio-Rad, USA). The cell proliferation inhibition rate was calculated relative to the DMSO-treated control group, and each experiment was performed in triplicate to ensure reproducibility.

Flow Cytometry Analysis for Cell Apoptosis

Flow cytometry was performed to detect the apoptotic rate of SKOV3 cells after combined treatment with Erastin and paclitaxel using an Annexin V-FITC/PI Apoptosis Detection Kit (Beyotime, China). SKOV3 cells treated with 4 μ M Erastin plus 10 μ M paclitaxel for 48 h were harvested by trypsin digestion (EDTA-free trypsin) and washed twice with pre-cooled phosphate-buffered saline (PBS). The collected cells were resuspended in 195 μ L of Annexin V-FITC binding buffer, and then 5 μ L of Annexin V-FITC and 10 μ L of propidium iodide (PI) were added successively. The mixture was gently mixed and incubated for 15 min at room temperature in the dark. Apoptotic cells were immediately detected using a flow cytometer (BD FACSCalibur, USA), and the data were analyzed with FlowJo software (Tree Star, USA). Cells were divided into four quadrants: viable cells (Annexin V⁻/PI⁻), early apoptotic cells (Annexin V⁺/PI⁻), late apoptotic cells (Annexin V⁺/PI⁺) and necrotic cells (Annexin V⁻/PI⁺). The total apoptotic rate was calculated as the sum of early and late apoptotic cell rates, and each experiment was repeated three times.

Lipid Peroxidation and Cytotoxicity Assays

Lipid peroxidation was evaluated by measuring malondialdehyde (MDA) levels using the MDA Assay Kit (ab118970, Abcam). Cytotoxicity was simultaneously assessed via the release of lactate dehydrogenase (LDH) activity, employing the Cytotoxicity Detection Kit (LDH) (Beyotime Biotech, China).

Transcriptome Sequencing and Bioinformatics Analysis

Transcriptome sequencing was performed on SKOV3 cells from two groups (3 biological replicates for the paclitaxel monotherapy group, 3 biological replicates for the Erastin plus paclitaxel combination therapy group), with a total of 6 samples. Total RNA was extracted from the treated cells using TRIzol reagent (Invitrogen, USA), and RNA quality and concentration were assessed using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA) and 1% agarose gel electrophoresis. Qualified RNA samples were used for cDNA library construction, and high-throughput sequencing was performed on the Illumina Novaseq platform (Illumina, USA) following the manufacturer's protocols.

Raw FASTQ sequencing data were subjected to strict quality control, including removal of adapter sequences, low-quality reads (Q-value < 20) and reads containing more than 5% unknown bases, to obtain high-quality clean reads. The clean reads were aligned to the human reference genome (hg38) using the STAR software (v2.7.1a) with default parameters for sequence alignment. Gene expression levels were quantified as raw read counts by HTSeq software (v0.13.5) based on the aligned results and human genome annotation file (GTF format). Differentially expressed genes (DEGs) between the two groups were identified using the DESeq2 R package (v1.38.0), with the screening criteria set as $|\log_2 \text{ fold change (FC)}| > 1$ and adjusted $P < 0.05$.

The DEGs were visualized by volcano plots, and the transcript expression correlation matrix of the 6 samples was calculated and plotted as a heatmap to evaluate intra-group repeatability and inter-group differences. For functional annotation and enrichment analysis, the clusterProfiler R package (v4.8.1) was used to perform Gene Ontology (GO) enrichment analysis, Reactome pathway analysis and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway

enrichment analysis of the DEGs. Significantly enriched pathways were screened with the criterion of $P < 0.05$ and visualized by bubble plots.

Statistical Analysis

All statistical analyses were performed using R software (version 4.1.2). Categorical clinicopathological variables were compared using the chi-square test, while continuous variables with non-normal distribution were analyzed with the Wilcoxon rank-sum test.³⁶ For all multiple enrichment and immune infiltration analyses, we uniformly adopted FDR correction to avoid inflation of type I error. Significant gene expression differences were identified using false discovery rate (FDR)-adjusted P -values.³⁷ Kaplan-Meier survival analysis with Log rank test was conducted to compare overall survival (OS) among subgroups.³⁸ The predictive accuracy of the prognostic model was evaluated by receiver operating characteristic (ROC) curve analysis using the “timeROC” package. Relationships between risk scores and immune cell infiltration were examined using Spearman correlation analysis. A P -value < 0.05 was considered statistically significant unless otherwise specified. All experiments were independently repeated at least three times, and all data were derived from biological replicates. Technical replicates were performed to ensure the reliability of experimental results.

Results

Ferroptosis Characteristics in Single-Cell Transcriptome

Single-cell RNA sequencing information from seven early-stage ovarian cancer patients was acquired from GEO database. After quality control, further cells from seven samples were included for downstream analysis. The Harmony R package was used to correct for batch effects. The top 2,000 highly variable genes were selected for dimensionality reduction and clustering at a resolution of 0.4, yielding 8 distinct cell clusters. Cell populations were annotated as B cells, endothelial cells, Epithelial cells, fibroblasts, myeloid cells, T cells and etc (Figure 1A). A scatter plot displayed the expression patterns of canonical marker genes for each cell type (Figure 1B). Ferroptosis-related gene set scores were calculated for all cells using the “AddModuleScore” function in Seurat (Figure 1C). Elevated ferroptosis activity was observed in Epithelial cells, myeloid cells, fibroblasts compared to other cell types (Figure 1D).

Identification of Ferroptosis-Related Core Modules and Genes in Bulk RNA Sequencing

The single-sample gene set enrichment analysis (ssGSEA) algorithm was used to compute ferroptosis activity scores for each sample in the TCGA-OV (The Cancer Genome Atlas—Ovarian Cancer) cohort. To identify gene modules significantly associated with ferroptosis activity, weighted gene co-expression network analysis (WGCNA) was performed on the same dataset. After removing outlier samples, a co-expression network was built using the 1,145 ferroptosis-related differentially expressed genes (DEGs) previously derived from single-cell RNA sequencing (Figure 2A). With a minimum module size of 50 and a module dissimilarity threshold (MEDissThres) of 0.15, five distinct co-expression modules were identified (Figure 2B). Correlation analysis showed that the blue module had a strong positive correlation with ferroptosis activity (correlation coefficient = 0.68; Figure 2C). A scatter plot of gene significance (GS) against module membership (MM) for genes in the blue module indicated a highly significant positive correlation (correlation coefficient = 0.77, $P = 9.2 \times 10^{-67}$; Figure 2D), implying that these genes may play central roles in ferroptosis regulation. A volcano plot displays DEGs between tumor and normal ovarian tissues from the TCGA-OV bulk RNA-seq data, filtered using $|\log FC| > 0.5$ and adjusted P value < 0.05 (Figure 2E). Intersection of the 334 genes in the blue module with the bulk RNA-seq DEGs yielded 223 overlapping genes (Figure 2F), which were thus identified as consistent ferroptosis-associated candidates across single-cell and bulk transcriptomic levels. Different from previously reported ferroptosis signatures merely constructed based on single-dataset bulk transcriptomic data, these candidate genes were screened via integrated single-cell and bulk omics validation, forming the multi-omics basis of our signature with incremental research novelty. Gene Ontology (GO) enrichment analysis of these ferroptosis-related genes revealed significant enrichments: under Biological Process (BP), terms included amide binding, peptide binding, and cytokine activity; under Cellular Component (CC), terms involved endocytic vesicles, vesicle lumens,

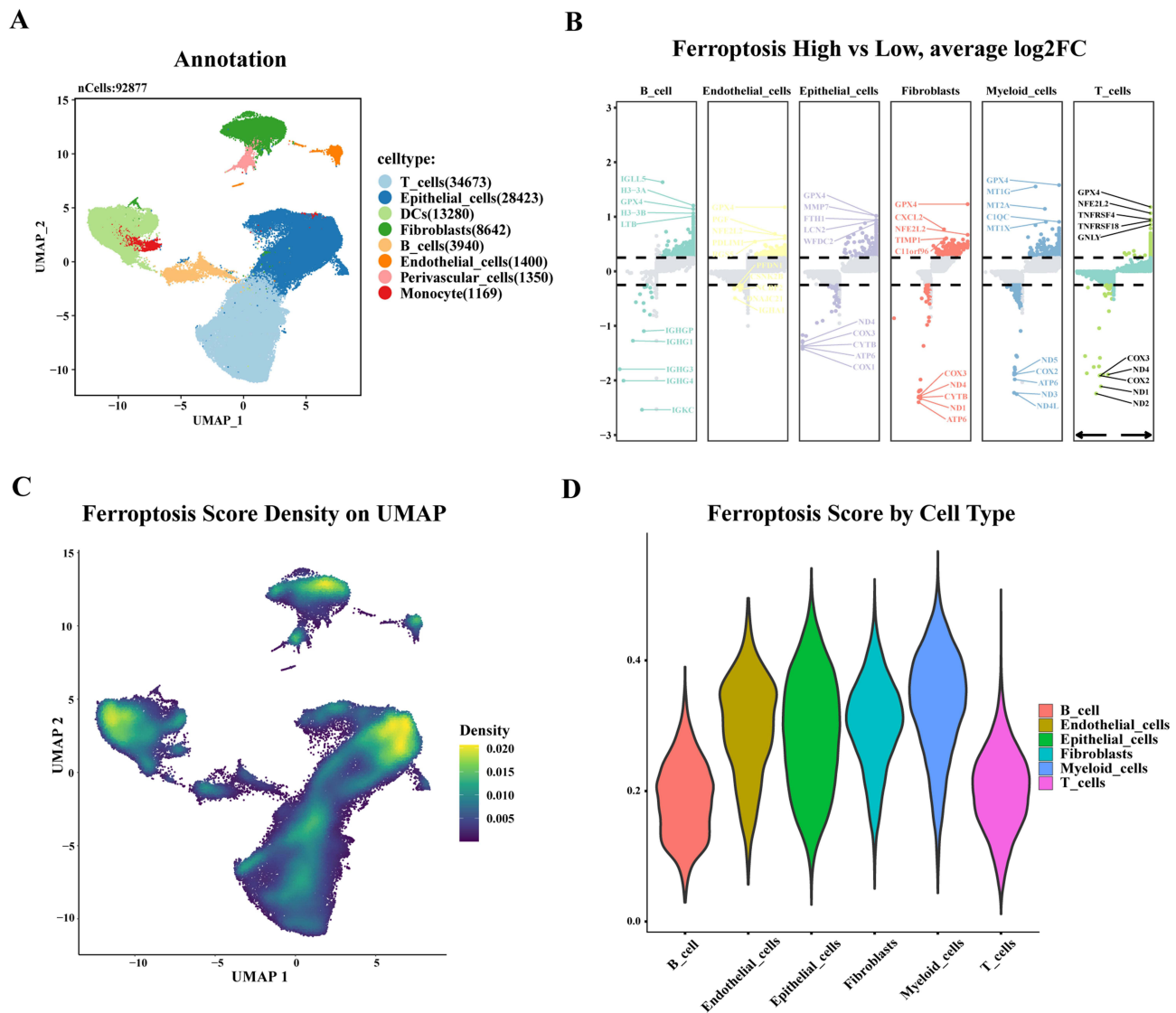


Figure 1 Ferroptosis-related signature (FRS) features in the single-cell transcriptome. **(A)** Cell clusters and major cell type annotation in ovarian cancer. **(B)** The top marker genes in each cell cluster. **(C)** Spatial distribution of ferroptosis score density on UMAP. **(D)** Ferroptosis score distribution across distinct cell types.

and cytoplasmic vesicle lumens; and under Molecular Function (MF), terms encompassed T-cell activation, neutrophil activation involved in immune response, and neutrophil degranulation (Figure 2G). Subsequently, univariate Cox regression analysis applied to the 223 ferroptosis-related genes identified 17 genes significantly linked to prognosis (Figure 2H). Copy number variation (CNV) analysis further indicated that the *BTN3A2* gene displayed a frequency of copy number gain exceeding 20% (Figure 2I).

Development and Validation of an Integrated Machine Learning-Based Prognostic Signature

A generalized correlation signature was constructed by applying 101 distinct machine learning algorithms to analyze 17 prognostic genes identified through univariate Cox regression. Compared with existing ovarian cancer ferroptosis signatures adopting only one or two machine learning models, this multi-algorithm screening framework improves the stability and avoids model selection bias, which constitutes another incremental innovation of this study. The model was trained on the GSE14764 dataset and externally validated using TCGA-OV data under a 10-fold cross-validation framework, with predictive performance assessed by concordance indices (C-indices) in both cohorts (Figure 3A).

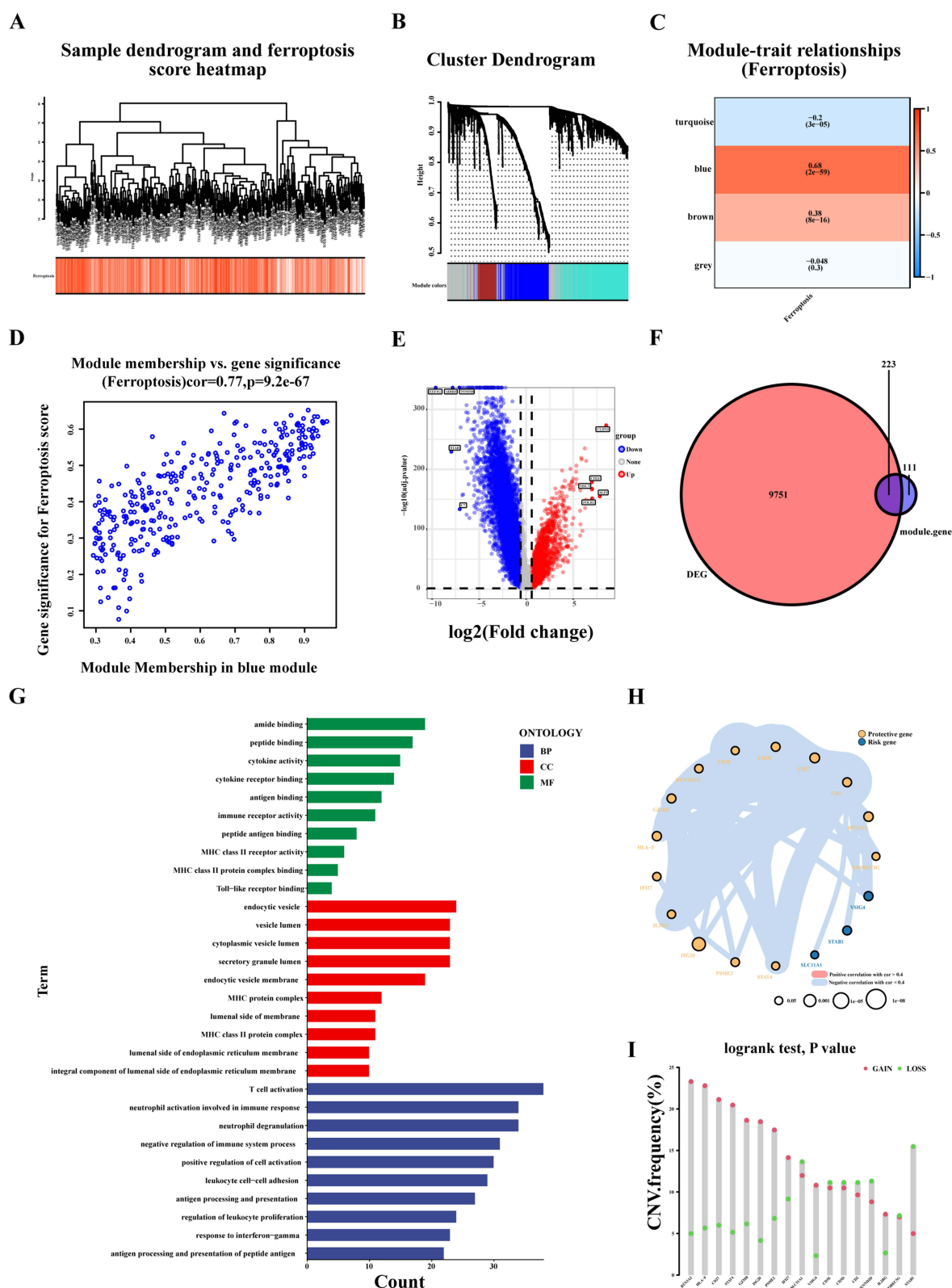


Figure 2 The ferroptosis-associated gene network and screening of key ferroptosis genes in ovarian cancer. **(A)** Sample clustering and ferroptosis score heatmap of ovarian cancer samples. **(B)** Co-expression gene modules identified by WGCNA. **(C)** Correlation between co-expression modules and ferroptosis activity. **(D)** Strong positive correlation between module membership and gene significance for ferroptosis in the blue module. **(E)** Differentially expressed genes between tumor and adjacent normal tissues. **(F)** Identification of core ferroptosis-related genes via intersection of the blue module and DEGs. **(G)** GO functional enrichment of core ferroptosis genes. **(H)** Protein-protein interaction network of core ferroptosis genes. **(I)** CNV alteration frequency of core ferroptosis genes.

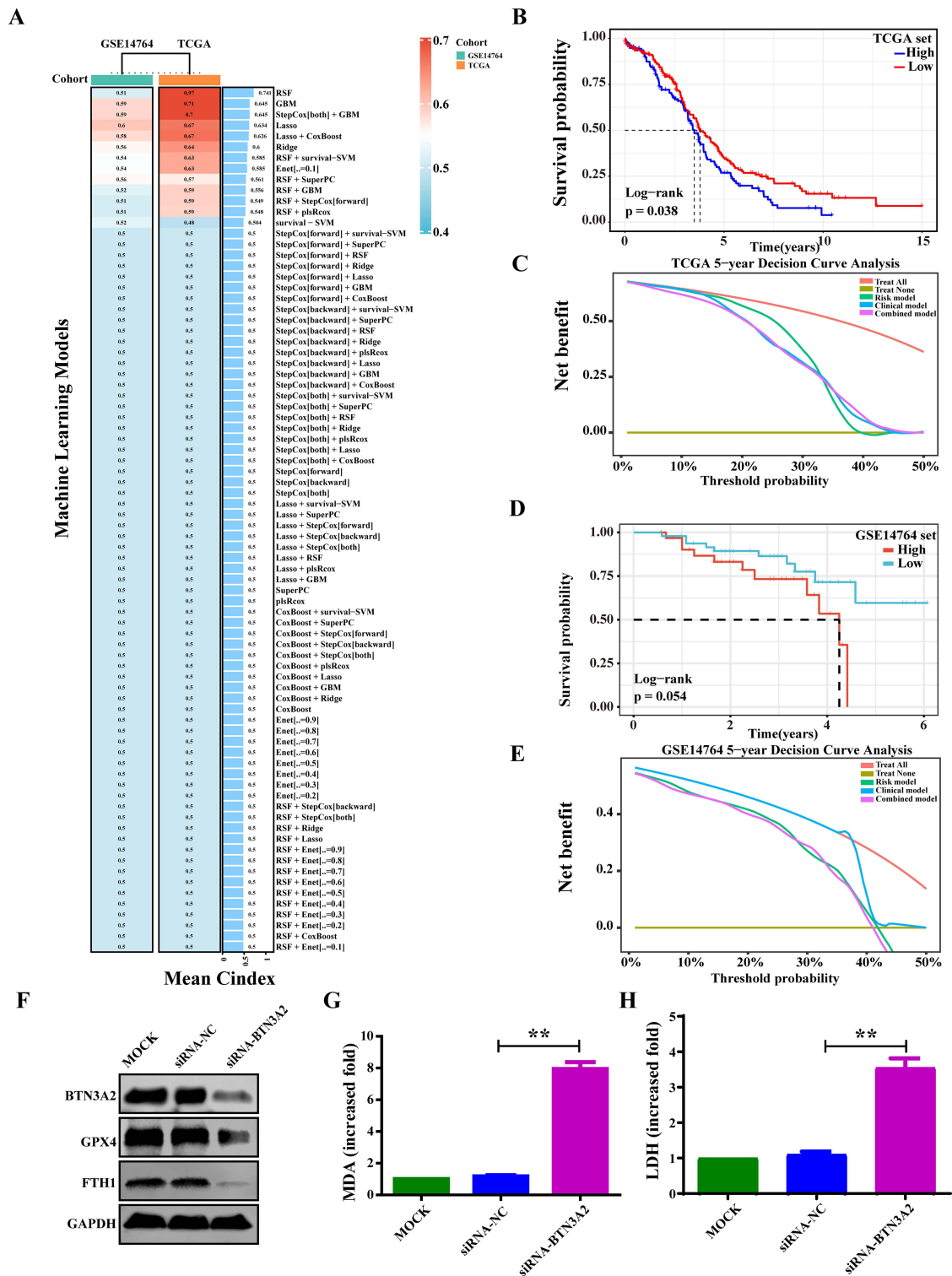


Figure 3 Construction and validation of the ovarian cancer prognostic model based on machine learning. **(A)** Heatmap of the concordance index (C-index) for machine learning models constructed in the GSE14764 and TCGA-OV cohorts. **(B)** KM survival curves for prognosis stratification in the TCGA-OV cohort. **(C)** 5-year decision curve analysis (DCA) of the combined nomogram model in the TCGA-OV cohort, comparing the net clinical benefit of treat-all, treat-none, clinical variables alone, risk signature alone, and integrated nomogram. **(D)** External validation KM survival curves in the GSE14764 cohort. **(E)** 5-year decision curve analysis (DCA) of the combined nomogram model in the GSE14764 training cohort, comparing the net clinical benefit of treat-all, treat-none, clinical variables alone, risk signature alone, and integrated nomogram. **(F)** Western blot analysis of BTN3A2, GPX4, FTH1, and GAPDH (loading control) protein expression levels in MOCK, siRNA-NC, and siRNA-BTN3A2 groups. **(G)** Quantitative analysis of intracellular malondialdehyde (MDA) content, indicating the level of lipid peroxidation. **(H)** Quantitative analysis of lactate dehydrogenase (LDH) release activity in the culture supernatant, reflecting cytotoxicity and membrane integrity. Data are presented as mean $\pm$ standard deviation (SD) from three independent experiments ($n=3$). ** $P < 0.01$, *** $P < 0.001$ versus the siRNA-NC group (one-way ANOVA).

Among the 101 models evaluated, “StepCox[both]+GBM” and “Lasso+CoxBoost” were selected due to their consistently robust performance across training and validation sets. In the GSE14764 training cohort, high-risk patients showed a trend toward shorter overall survival compared to low-risk patients ($P=0.054$, [Figure 3D](#)). This prognostic distinction was confirmed in the TCGA-OV cohort, where high-risk patients had significantly poorer survival outcomes ($P=0.038$, [Figure 3B](#)). To quantitatively evaluate the clinical net benefit of our signature-integrated nomogram, we performed 5-year decision curve analysis (DCA) for both training and validation cohorts. The DCA curves for the TCGA cohort and GSE14764 cohort are displayed in [Figure 3C](#) and [E](#), respectively. Across a wide range of threshold probabilities, the combined nomogram incorporating the ferroptosis risk signature yielded superior net clinical benefit compared with the clinical-only model and risk signature alone in both datasets.

Among the 17 prognostic genes prioritized from our initial screening, *BTN3A2* captured our particular interest. To investigate its functional role in SKOV ovarian cancer cells, we knocked down *BTN3A2* using small interfering RNA (siRNA) and verified the knockdown efficiency via Western blotting. As shown in [Figure 3F](#), the protein level of *BTN3A2* was markedly reduced in the siRNA-*BTN3A2* group compared to the MOCK and siRNA-NC groups. Given the potential link between *BTN3A2* and ferroptosis, we subsequently examined the expression of two critical ferroptosis-related markers, GPX4 and FTH1. The results demonstrated that *BTN3A2* knockdown significantly downregulated both GPX4 and FTH1 protein levels. Accordingly, we further measured the cellular lipid peroxidation level and cytotoxicity. As presented in [Figure 3G](#) and [H](#), *BTN3A2* depletion significantly increased the intracellular malondialdehyde (MDA) content and the release of lactate dehydrogenase (LDH) activity into the culture supernatant. Collectively, these findings suggest that *BTN3A2* deficiency promotes lipid peroxidation and exacerbates cell membrane damage in SKOV3 cells.

Time-dependent ROC analysis yielded AUC values of 0.72, 0.75, and 0.77 for 1-, 2-, and 3-year survival, respectively, in the training set. Corresponding AUC values in the validation set were 0.57, 0.67, and 0.74 ([Figure S1A](#) and [B](#)). Notably, the 1-year AUC of only 0.57 in the independent validation set indicated limited predictive power for short-term survival outcomes, which represents an important limitation of this ferroptosis-related signature. These moderate AUC values suggested that this FRS serves as an exploratory prognostic biomarker rather than a mature clinical diagnostic tool for routine clinical application. Given the clinical relevance of standard prognostic parameters in ovarian cancer, we further investigated the relationship between the ferroptosis-related signature and clinicopathological variables ([Figure S1C](#)). While no significant difference in risk scores was observed between stage III–IV and stage I–II patients ([Figure S1D](#)), the distribution of survival status in TCGA-OV markedly differed between risk subgroups ([Figure S1E](#)). The signature also exhibited potential predictive value for tumor staging ([Figure S1F](#)). Kaplan-Meier analysis confirmed that high-risk patients had significantly shorter survival across all clinical stages ($P=0.017$), supporting the robust prognostic value of the ferroptosis-related model ([Figure S1G](#)). Diagnostic ROC analysis showed AUCs of 0.534, 0.516, and 0.565 for the signature in predicting tumor staging ([Figure S1H–J](#)), suggesting its potential utility in identifying metastatic progression. Cox regression analyses in both TCGA-OV and GSE14764 confirmed the FRS as an independent prognostic factor for OS in univariate and multivariate models ([Figure S2A](#) and [B](#)). To enhance clinical applicability, a nomogram integrating the FRS with key clinical variables was developed ([Figure S2C](#)) and validated using calibration curves ([Figure S2D](#)). The nomogram quantified individual 1-, 3-, and 5-year OS probabilities by summing points of RiskScore and tumor stage ([Figure S2E](#)).

Molecular Mechanisms Associated with the Ferroptosis-Related Signature (FRS) in Bulk Transcriptome Data

To elucidate the molecular mechanisms through which the ferroptosis-related signature (FRS) influences ovarian cancer prognosis, gene set enrichment analysis (GSEA) was conducted. Using the “HALLMARK” gene set, we found that the low-risk group was primarily enriched in pathways including MYC targets V1, E2F targets, and oxidative phosphorylation ([Figure 4A](#)). In contrast, the high-risk group showed marked enrichment in HALLMARK_NOTCH_SIGNALING, TNFA signaling via NF- κ B, and apical junction pathways ([Figure 4B](#)).

Gene set variation analysis (GSVA) based on the “HALLMARK” collection further revealed distinct pathway activities between risk groups. The high-risk group displayed enhanced activation of apical junction, apical surface, and NOTCH signaling, whereas the low-risk group exhibited higher activity in MYC targets V1, spermatogenesis, and

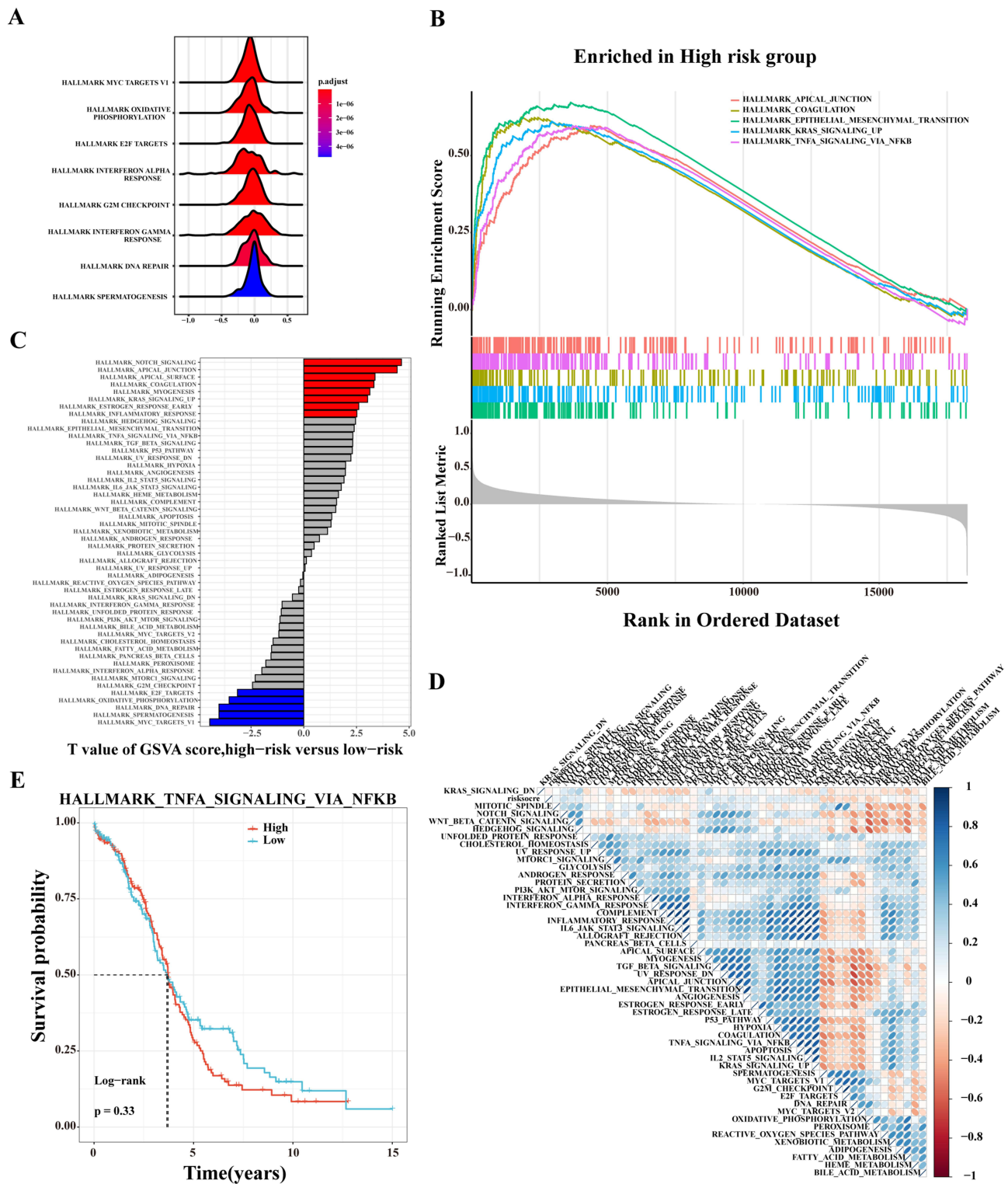


Figure 4 Pathway activity and survival analysis of high- and low-risk groups. **(A)** Ridgeline plot of HALLMARK pathway activity. **(B)** GSEA enrichment analysis of pathways in the high-risk group. **(C)** GSVA pathway activity scores between risk groups. **(D)** Correlation heatmap of HALLMARK pathways and FRS risk scores. **(E)** Kaplan-Meier survival curves for the TNFA_SIGNALING_VIA_NFKB pathway.

DNA repair pathways (Figure 4C). Correlation analysis between FRS scores and hallmark pathway activities reinforced these patterns (Figure 4D), indicating that the FRS is closely linked to multiple cancer-related biological processes and metabolic pathways. To further evaluate the clinical significance of these pathways, we conducted Kaplan-Meier survival

analysis. Pathways that exhibited a positive correlation with the FRS, such as TNFA signaling via NF- κ B and hypoxia, were linked to poorer prognoses. In contrast, pathways that showed a negative correlation, including cholesterol homeostasis (Figure 4E), were associated with more favorable outcomes. Collectively, these findings indicate that targeting critical ferroptosis-associated pathways may provide new therapeutic opportunities for ovarian cancer.

Genomic Alterations and Tumor Heterogeneity Among Ferroptosis-Related Signature (FRS) Subgroups

Analysis revealed that elevated mutant-allele tumor heterogeneity (MATH) scores were not strongly associated with poor prognosis in ovarian cancer patients, and no significant difference in overall survival (OS) was observed between patients with high versus low MATH scores (Figure 5A and B). However, when combining intratumor heterogeneity with the ferroptosis-related signature (FRS), patients in the “high-risk, high-MATH” subgroup exhibited significantly worse survival compared to the “low-risk, low-MATH” subgroup (Log rank test, $P=0.016$), indicating that integrating these two parameters may improve prognostic stratification in ovarian cancer (Figure 5C). To investigate genomic alterations between FRS-defined subgroups, mutation profiles were compared. The USH2A gene, which is associated with cancer progression via the regulation of gene expression, signaling, and cell proliferation, exhibited a higher mutation frequency in the high-risk group (11%) compared to the low-risk group (6%). Moreover, the tumor suppressor NF1, a regulator of RAS signaling, was mutated in 8% of high-risk patients but not detected in the low-risk group (Figure 5D–E). Co-occurrence and mutual exclusivity patterns among the top 20 mutated genes were also examined, revealing more frequent co-occurring mutations in the high-risk group (Figure 5F and G). Copy number variation (CNV) analysis of the 15 most differentially altered genes between risk subgroups identified prevalent copy number gains in *SI*, *CSMD3*, *FLG*, and *RYR2*, while deletions were more common in *SYNE2*, *ARHGAP35*, and *NF1* (Figure 5H). These findings indicate that CNVs may significantly influence gene expression and contribute to molecular heterogeneity in ovarian cancer.

Association Between the Ferroptosis-Related Signature (FRS) and Biological Characteristics

To investigate the potential influence of the ferroptosis-related signature (FRS) on the tumor microenvironment (TME), we analyzed single-cell transcriptomic data to characterize the expression patterns of *IFI27*, *BTN3A2*, *DENND2D*, and *HLA-F* across different cell types (Figure 6A). Functional enrichment analysis indicated predominant involvement of these genes in ribosome biogenesis, cell adhesion molecules, and phagosome-related pathways (Figure 6B). Gene Set Enrichment Analysis (GSEA) using the “HALLMARK” gene set showed that high-risk group genes were mainly enriched in interferon-alpha response, interferon-gamma response, and TNFA signaling via NF- κ B pathways (Figure 6C). It should be acknowledged that the intrinsic mechanistic links between these four core signature genes (*IFI27*, *BTN3A2*, *DENND2D*, *HLA-F*) and ferroptosis biological processes remain purely supported by bioinformatic correlation analysis in the current study; we have not yet performed targeted cellular or molecular functional experiments to validate their direct regulatory roles in ferroptosis signaling cascades.

We further stratified tumor cells by FRS scores to examine their communication with other TME components. Distinct intercellular interaction patterns were observed between high- and low-risk tumor cells (Figure 6D and E). Within the TME, cells functioned as “senders,” “receivers,” “mediators,” or “influencers” in specific signaling networks. High-risk tumor cells demonstrated more extensive and complex crosstalk with other TME cells, particularly through pathways involving intercellular adhesion molecules (ICAM), vascular endothelial growth factor (VEGF), and T-cell immunoglobulin and ITIM domain (TIGIT) signaling. These broad communication networks and their strong mediating effects suggest that high-risk tumor cells can remodel surrounding cell behavior, influencing processes such as cell adhesion, angiogenesis, immune evasion, and ultimately tumor survival (Figure 6F–H). Using the ESTIMATE algorithm, we evaluated immune infiltration patterns across FRS-based risk groups. The high-risk group showed significantly elevated immune, stromal, and ESTIMATE scores, along with higher tumor purity, whereas the low-risk group displayed lower values across these metrics (Figure S3A–C). Notably, the high-risk subgroup presents an inflamed but functionally exhausted immunosuppressive TME phenotype: increased immune cell infiltration, elevated immune scores and stromal

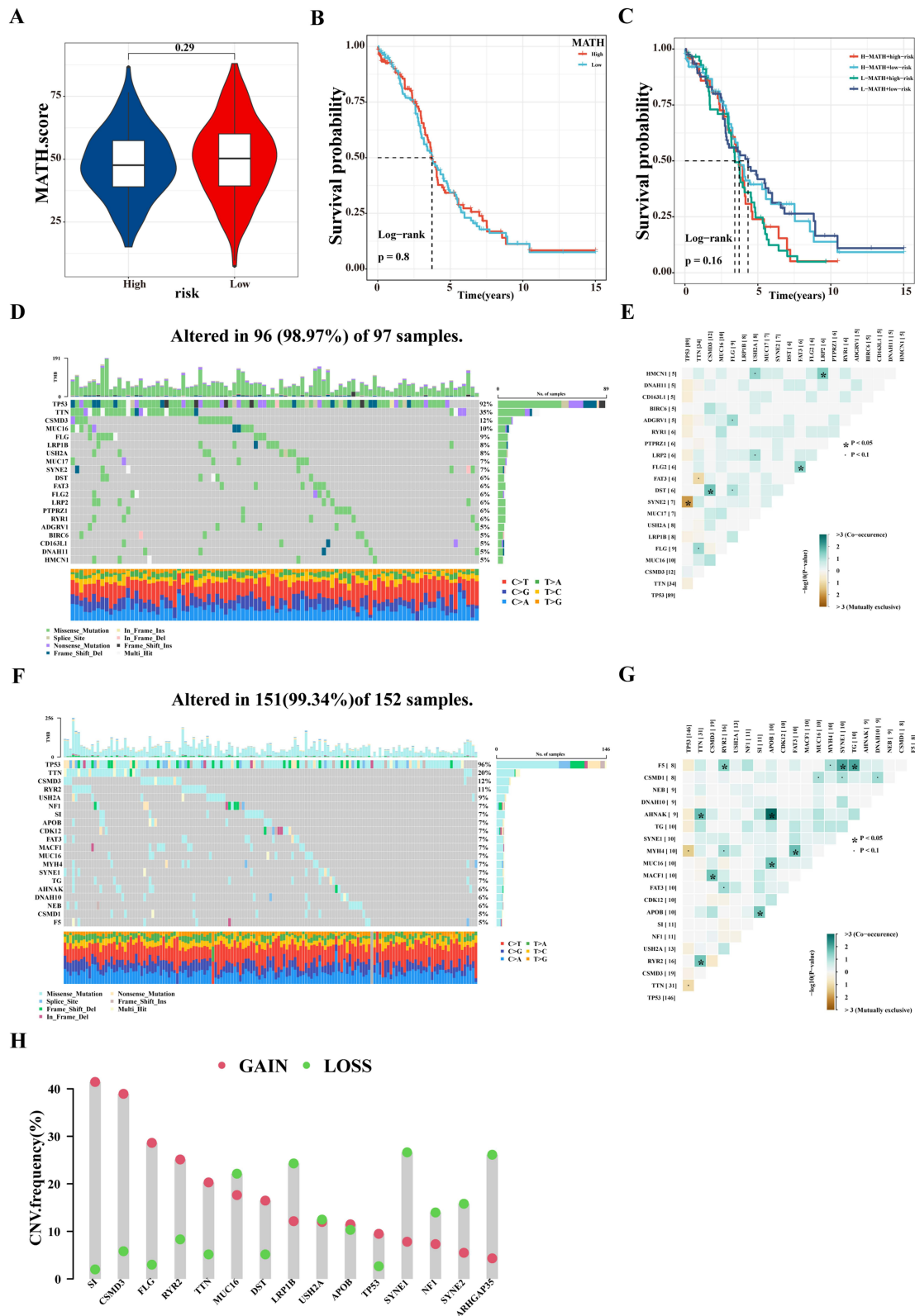


Figure 5 Genetic alterations associated with FRS between low- and high-risk groups. **(A)** Violin plot of MATH scores between high-risk and low-risk groups. **(B)** Kaplan-Meier survival curves for high vs. low MATH score groups. **(C)** Kaplan-Meier survival curves combining MATH scores and FRS risk stratification. **(D and E)** Waterfall plots of somatic mutation landscapes in low-risk **(D)** and high-risk **(E)** groups. **(F and G)** Heatmaps of co-occurrence and mutual exclusivity among top mutated genes in low-risk **(F)** and high-risk **(G)** groups. **(H)** CNV frequency distribution of genes between high-risk and low-risk groups, with gains (red) and losses (green).

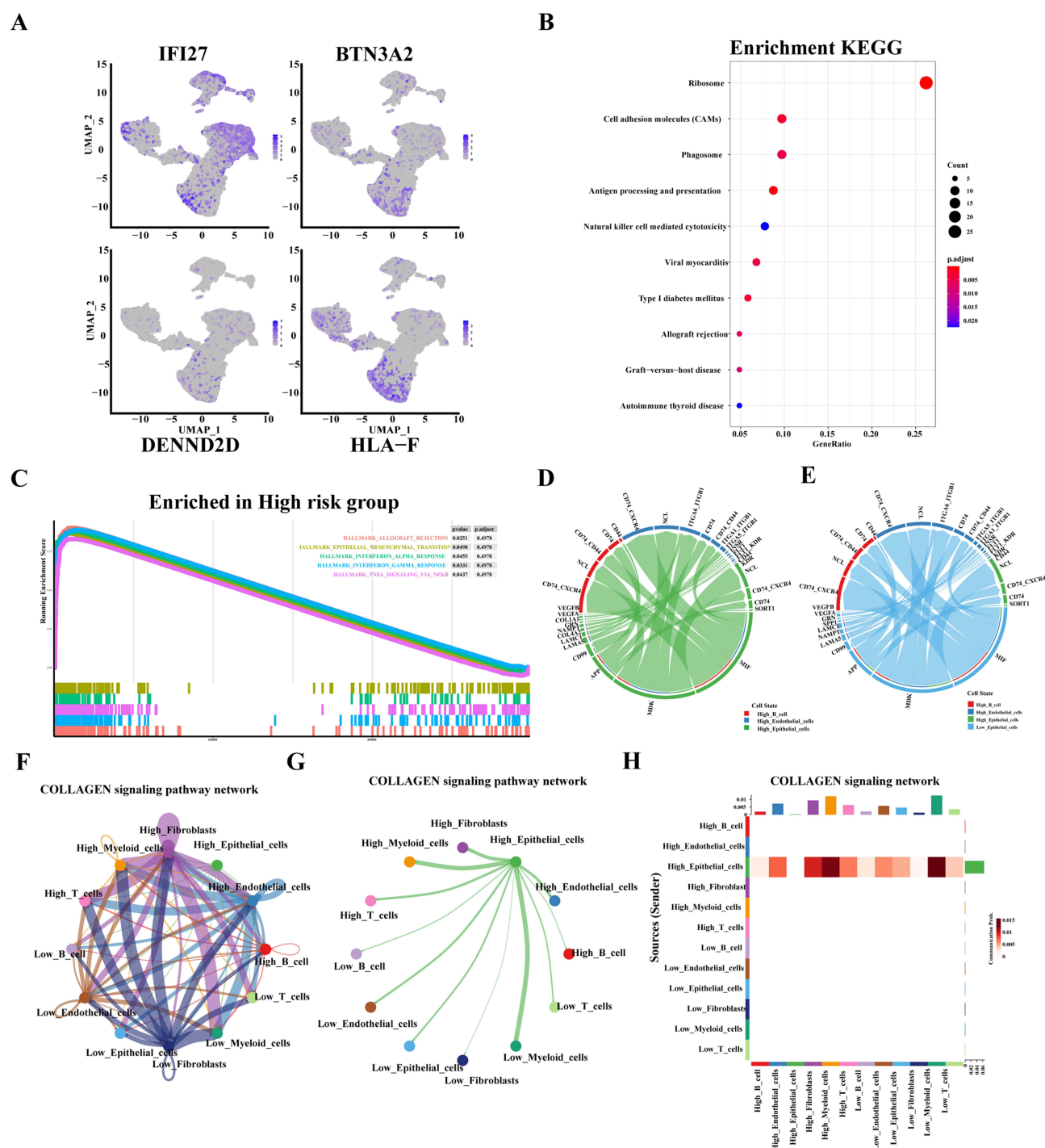


Figure 6 Regulatory mechanisms of ferroptosis-related pathways and cellular communication networks in ovarian cancer. **(A)** Spatial expression patterns of feature genes on UMAP. **(B)** KEGG enrichment analysis of differential pathways between risk groups. **(C)** GSEA enrichment analysis of pathways in the high-risk group. **(D and E)** Circular heatmaps of intercellular communication strength in high-risk **(D)** and low-risk **(E)** groups. **(F)** Collagen signaling cell-cell communication network in high- and low-risk groups. **(G)** Collagen signaling pathway network with sender and receiver cell roles. **(H)** Heatmap of cell roles in the collagen signaling pathway network.

scores reflect activated immune response, while high checkpoint expression and disordered intercellular crosstalk inhibit anti-tumor immune function, eventually forming immune exhaustion status, which explains the contradictory immune features in high-risk population. Pathway activity analysis indicated a marked activation of immune-related pathways in the high-risk group, including Toll-like receptor (TLR) signaling, natural killer (NK) cell-mediated cytotoxicity, and chemokine signaling (Figure S3D). CIBERSORT-based immune cell quantification (Figure S3E) indicated that the high-

risk group contained increased proportions of activated CD4⁺ memory T cells, regulatory T cells (Tregs), CD56⁺ natural killer cells, and plasma cells. In contrast, the low-risk group was enriched for resting CD4⁺ memory T cells, M2 macrophages, and resting mast cells ([Figure S3F](#)). Correlation analysis of 15 FRS-associated genes with immune infiltration demonstrated significant associations; for instance, CD3E exhibited a positive correlation with CD8⁺ T cells, BLOC1S1 correlated with $\gamma\delta$ T cells, and GZMB was associated with M1 macrophages ([Figure S3G](#)). Spearman analysis identified 22 immune cell types significantly linked to the FRS ([Figure S3H](#)). Integrating ANOVA, correlation, and survival analyses, we ultimately identified five key TME cell types as pivotal determinants of ovarian cancer prognosis and progression ([Figure S3I](#)).

Relationship Between the Ferroptosis-Related Signature (FRS), Cancer Immunity Cycle, Immunotherapy Response, and Drug Sensitivity

Recent advances in targeted therapy, particularly with PARP inhibitors, have demonstrated substantial clinical progress in ovarian cancer treatment. The evolving molecular classification of ovarian cancer has revealed its genetic heterogeneity, characterized by diverse driver mutations and genomic alterations, thereby creating new opportunities for targeted interventions. However, the dynamic and heterogeneous nature of the tumor microenvironment (TME) continues to pose a major clinical challenge through the development of therapy resistance.⁴ To explore this further, we evaluated the sensitivity of FRS-defined risk subgroups to five anticancer agents. Analysis of half-maximal inhibitory concentration (IC50) values showed that the low-risk subgroup had significantly increased sensitivity to camptothecin ([Figure S4B](#)), with risk scores positively correlating with camptothecin sensitivity ([Figure S4G](#)). Conversely, the high-risk subgroup demonstrated enhanced sensitivity to A.443654, BMS.708163, etoposide, and PF.4708671 ([Figure S4A](#) and [C–E](#)), as evidenced by negative correlations between risk scores and IC50 values for these agents ([Figure S4F](#) and [H–J](#)). These results indicate that low-risk patients may derive greater benefit from camptothecin treatment, whereas high-risk patients appear more responsive to CHIR-99021, imatinib, LFM-A13, or bexarotene. Together, these findings position the ferroptosis-related signature as a valuable predictor of drug sensitivity in ovarian cancer, with potential utility in guiding personalized treatment strategies.

Association Between the Ferroptosis-Related Signature (FRS) and Immune Microenvironment and Therapeutic Response

To explore the clinical and biological relevance of the ferroptosis-related signature (FRS) in ovarian cancer, we first characterized the landscape of immune cell infiltration in the tumor microenvironment (TME) based on FRS risk stratification ([Figure 7A](#)). The high-risk group was characterized by enriched infiltration of CD8⁺ effector T cells, monocytes.CD14, monocytes.CD16, and plasma cells, while the low-risk group showed higher proportions of CD4⁺ naive T cells, CD4⁺ memory T cells, and natural killer (NK) cells, suggesting profound remodeling of the immune microenvironment by the FRS. We further evaluated the expression levels of key immune checkpoint molecules (ICMs) and immune-related genes between the risk groups ([Figure 7B](#)). Quantitative analysis demonstrated that the high-risk group exhibited significantly higher expression levels of immune checkpoint factors including CD27, CTLA4, PDCD1, LAG3, TIGIT, TNFSF14, and TNFSF25 compared to the low-risk group. These findings indicate that the high-risk FRS profile may drive immune suppression by upregulating critical immune checkpoint signaling, thereby facilitating immune evasion in the TME. To validate the immune regulatory potential of FRS, we performed subgroup analyses focusing on immune checkpoint signature scores ([Figure 7C–E](#)). The high-risk group displayed significantly elevated *ips_cta4_neg_pdl1_neg*, *ips_cta4_neg_pdl1_pos* scores, and higher tumor mutation burden (TMB) levels. These observations suggest that high-FRS tumors possess a more active immune escape phenotype and higher genomic instability, which may contribute to their aggressive biological behavior. We finally explored the correlation between FRS and clinical therapeutic response, using treatment outcomes (complete response/partial response (CR/PR), stable disease (SD), and progressive disease (PD)) as stratification criteria ([Figure 7F–H](#)). The risk score distribution varied significantly across different response groups. Patients achieving CR/PR had the lowest risk scores, while those with PD exhibited the highest risk scores. Additionally, the proportion of patients with favorable treatment response (CR/PR) was markedly lower in the high-risk subgroup, whereas the low-risk subgroup was enriched for treatment-responsive individuals. Collectively, these results demonstrate that the FRS is closely associated with the immunosuppressive

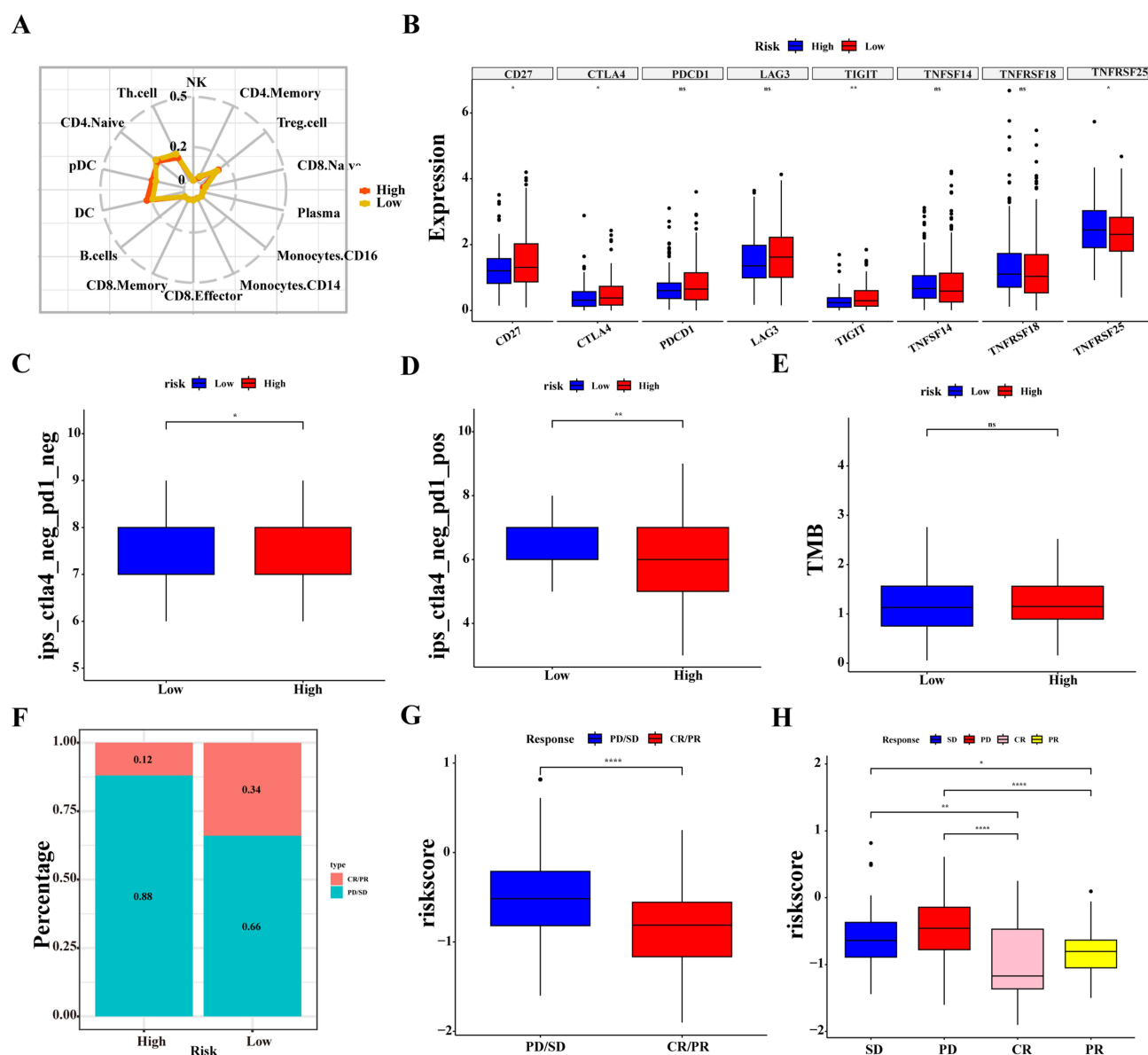


Figure 7 Relationship between FRS and Immunotherapy. **(A)** Radar graph showing differences in the ability of FRS subgroups to recruit immune cells. **(B)** Immune checkpoint expression in high and low FRS risk subgroups. **Statistical significance markers: *P < 0.05, **P < 0.01, *P < 0.001, ns, no significance. **(C)** IPS scores for ips_ctla4_neg_pd1_neg (CTLA4-/PD-1-treatment) between the high and low-risk groups. **(D)** IPS scores for ips_ctla4_neg_pd1_pos (CTLA4-/PD-1+ treatment) between the high- and low-risk groups. **(E)** Box plot showing changes in tumor mutational burden (TMB) scores between the high- and low-risk groups in the IMvigor210 cohort. **Statistical significance markers: *P < 0.05, **P < 0.01, *P < 0.001, ns, no significance. **(F)** Proportion of patients with CR/PR or PD/SD receiving immunotherapy in the high-risk and low-risk groups of the IMvigor 210 cohort. **(G)** Box plots depicting the difference in risk scores between patients with CR/PR and patients with PD/SD in the IMvigor 210 cohort. **Statistical significance markers: *P < 0.05, **P < 0.01, *P < 0.001, ns, no significance. **(H)** Box plot showing changes in risk scores for CR, PR, SD, and PD patients in the IMvigor210 cohort. **Statistical significance markers: *P < 0.05, **P < 0.01, *P < 0.001, ns, no significance.

TME and can serve as a reliable predictor of therapeutic response in ovarian cancer, providing a potential biomarker for personalized immunotherapy strategies.

Synergistic Anti-Tumor Effect of Erastin-Paclitaxel Combination in Ovarian Cancer Cells

To investigate the anti-tumor effects of the ferroptosis inducer Erastin combined with paclitaxel on ovarian cancer cells, we performed a series of in vitro experiments on two ovarian cancer cell lines including SKOV3 and Caov-3 to improve experimental reliability. First, SKOV3 cells were treated with serial concentrations of Erastin, and Western blot analysis was conducted to detect the expression of ferroptosis-associated proteins SLC7A11, GPX4 and FTH1 (Figure 8A). With

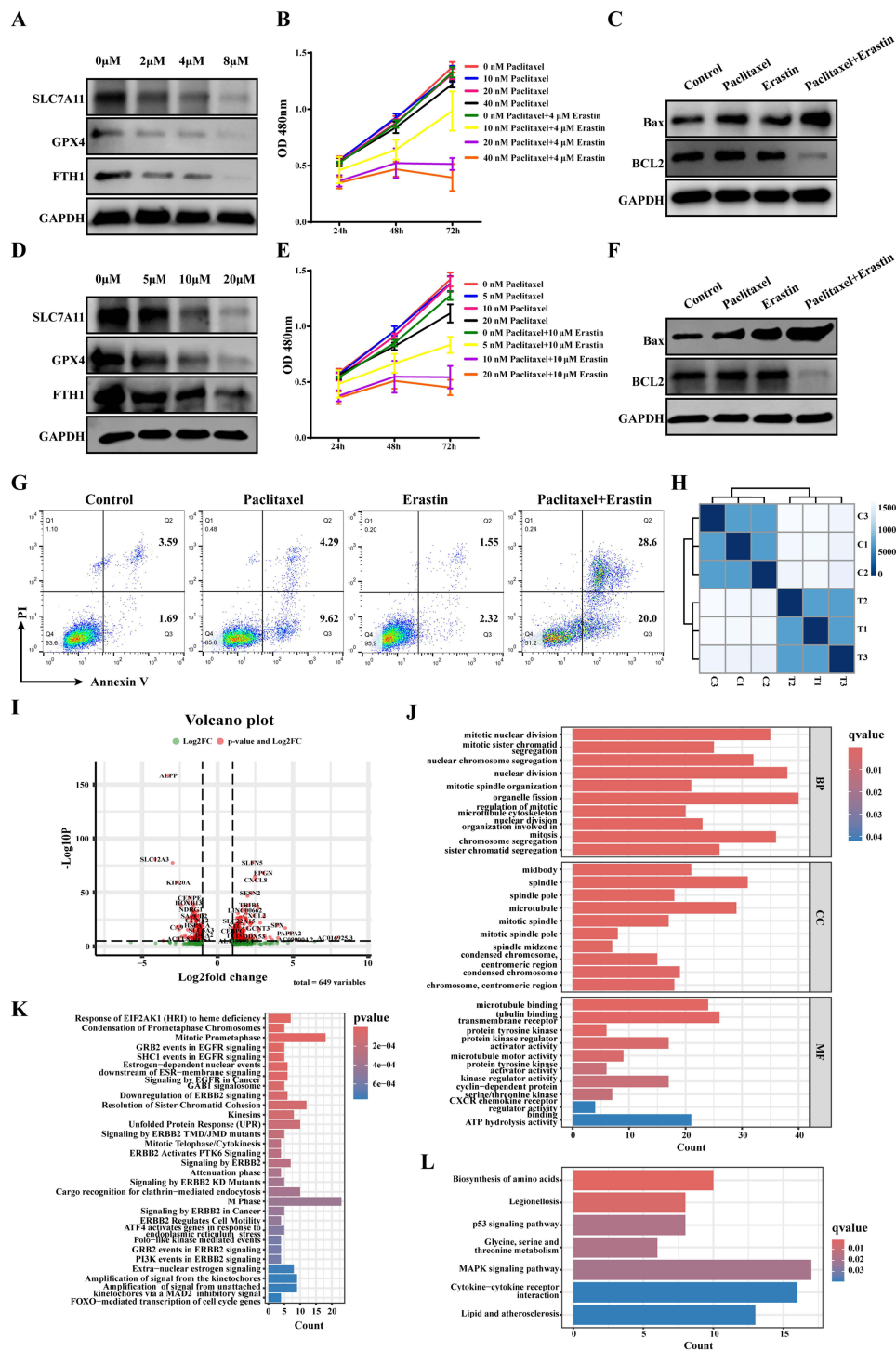


Figure 8 Anti-Tumor Effects of Erastin Combined with Paclitaxel on Ovarian Cancer Cells. (A) SKOV3 cells were treated with serial concentrations of the ferroptosis inducer Erastin, followed by Western blot analysis to detect the expression of ferroptosis-associated proteins SLC7A11, GPX4, and FTH1. (B) MTT assay was performed to evaluate the effect of paclitaxel at varying concentrations in combination with the ferroptosis inducer Erastin on the proliferation of SKOV3 cells. (C) SKOV3 cells were treated with 4 μ M Erastin in combination with 10 μ M paclitaxel, and Western blot was conducted to examine the expression of apoptosis-related proteins Bax and BCL2. (D) Caov-3 cells were treated with serial concentrations of the ferroptosis inducer Erastin, followed by Western blot analysis to detect the expression of ferroptosis-associated proteins SLC7A11, GPX4, and FTH1. (E) MTT assay was performed to evaluate the effect of paclitaxel at varying concentrations in combination with the ferroptosis inducer Erastin on the proliferation of Caov-3 cells. (F) Caov-3 cells were treated with 4 μ M Erastin in combination with 10 μ M paclitaxel, and Western blot was conducted to examine the expression of apoptosis-related proteins Bax and BCL2. (G) Flow cytometry was used to detect SKOV3 cells treated with 4 μ M Erastin in combination with 10 μ M paclitaxel. (H) Heatmap of the correlation matrix of transcript expression in 6 SKOV3 cell samples (3 control groups treated with paclitaxel alone, 3 treated with paclitaxel and ferroptosis inducer Erastin). (I) Volcano plot showing differentially expressed genes (DEGs) in SKOV3 cells treated with the combination therapy. (J) Bubble plot showing GO Analysis of enriched pathways in SKOV3 cells treated with the combination therapy. (K) Bubble plot showing Reactome Analysis of enriched pathways in SKOV3 cells treated with the combination therapy. (L) Bubble plot showing KEGG Analysis of enriched pathways in SKOV3 cells treated with the combination therapy.

the increase of Erastin concentration, the protein levels of SLC7A11, GPX4 and FTH1 were gradually decreased in a dose-dependent manner. MTT assay was then used to evaluate the effect of different concentrations of paclitaxel alone or in combination with Erastin on the proliferation of SKOV3 cells, and the results revealed an inherent resistance of SKOV3 cells to paclitaxel, while the combined treatment with Erastin markedly inhibited cell proliferation and promoted apoptotic cell death (Figure 8B). This pro-apoptotic effect was further verified by flow cytometry detection of SKOV3 cells treated with 4 μ M Erastin plus 10 μ M paclitaxel (Figure 8G). Western blot analysis of apoptosis-related proteins in SKOV3 cells after treatment with 4 μ M Erastin and 10 μ M paclitaxel showed a significant decrease in the expression of the anti-apoptotic protein BCL2 and a notable upregulation of the pro-apoptotic protein Bax (Figure 8C). To further validate our observations in paclitaxel-sensitive ovarian cancer cells, we replicated the functional experiments in Caov-3 cells. Gradient Erastin treatment followed by Western blot confirmed that Erastin also dose-dependently downregulated ferroptosis markers SLC7A11, GPX4 and FTH1 in Caov-3 cells (Figure 8D). MTT proliferation assays (Figure 8E) demonstrated that although Caov-3 cells exhibited intrinsic sensitivity to paclitaxel monotherapy, co-administration of Erastin generated stronger growth-suppressive effects. Consistently, Western blot for apoptotic proteins in Caov-3 cells (Figure 8F) showed that Erastin-paclitaxel co-treatment reduced anti-apoptotic BCL2 expression and increased pro-apoptotic Bax levels, mirroring the regulatory pattern observed in SKOV3 cells. To elucidate the molecular mechanisms underlying the pro-apoptotic effect of the combined therapy, transcriptome sequencing was performed on SKOV3 cells treated with paclitaxel alone or in combination with Erastin. The correlation matrix heatmap of transcript expression exhibited low intra-group variation in the two groups, indicating high repeatability of the sequencing data (Figure 8H). Volcano plot analysis of differentially expressed genes (DEGs) showed a relatively balanced distribution of upregulated and downregulated genes in the combined treatment group compared with the paclitaxel monotherapy group (Figure 8I). Functional enrichment analysis of these DEGs revealed that GO terms were mainly enriched in biological processes and cellular components related to cell division and nuclear function (Figure 8J). Reactome analysis indicated significant enrichment in cell cycle-related pathways including M phase and Mitotic Prometaphase (Figure 8K). KEGG pathway enrichment analysis further demonstrated that the DEGs were closely associated with the MAPK signaling pathway and lipid metabolism-related pathways (Figure 8L). Limited by the lack of ferroptosis rescue experiments, lipid ROS detection and quantitative synergy index verification, current cellular results are preliminary exploratory evidence. Collectively, these preliminary findings suggest that Erastin combined with paclitaxel may exert potential synergistic anti-tumor effects on ovarian cancer cells by regulating ferroptosis, apoptosis, cell cycle and multiple signaling pathways.

Discussion

Ovarian cancer is characterized by high mortality and poor prognosis, largely due to the lack of reliable prognostic biomarkers for effective risk stratification.³⁹ In this study, we constructed a ferroptosis-related signature (FRS) by integrating bulk transcriptomic and single-cell multi-omics data, combined with weighted gene co-expression network analysis (WGCNA) and multiple machine learning algorithms. This integrated approach overcomes the inherent drawbacks of single-omics analysis and greatly improves the robustness and population generalizability of the prognostic model. The FRS effectively stratified ovarian cancer patients into high-risk and low-risk subgroups with significant differences in overall survival, and this prognostic value was validated across multiple independent cohorts, including TCGA-OV and GSE14764. Univariate and multivariate Cox regression analyses further confirmed that FRS is an independent prognostic factor, independent of traditional clinicopathological parameters such as tumor stage, demonstrating its superior predictive value beyond conventional clinical indicators. Additionally, the nomogram constructed by integrating FRS with clinical characteristics significantly improved the accuracy of individualized survival prediction, and decision curve analysis verified its higher net clinical benefit compared to single clinical parameters. Existing mechanistic and clinical evidence supports the predictive capacity of FRS for immunotherapy and platinum chemotherapy response. Low-FRS tumors exhibit a “hot” immune microenvironment with abundant CD8⁺ T cells and elevated PD-L1, correlating with favorable ICB efficacy.⁴⁰ Platinum agents exert cytotoxicity by triggering ferroptosis, whereas suppressed ferroptosis (high FRS) drives platinum resistance.⁴¹ Notably, nearly all previously reported ferroptosis-related ovarian cancer signatures rely solely on retrospective TCGA/GEO cohorts to infer therapeutic response, as large-scale real-world cohorts paired with transcriptomic profiles and standardized treatment outcomes are scarce across the

field.^{42,43} Our bioinformatic results align with these published findings, validating FRS as a rational predictive marker. Notably, all analytical validation of this nomogram relies solely on retrospective public cohort data; formal prospective clinical cohort validation is still required to reliably confirm its real-world clinical utility. These findings are consistent with previous studies highlighting the prognostic significance of ferroptosis-related genes in various cancers, further supporting that aberrant ferroptosis regulation correlates statistically with ovarian cancer malignant progression and clinical prognosis.^{40,44} Notably, all transcriptomic and multi-omics analyses presented herein only establish correlational associations between ferroptosis signatures and clinical outcomes; causal mechanistic relationships cannot be definitively inferred solely from these computational data. Collectively, the FRS provides a reliable molecular tool for clinical prognostic stratification of ovarian cancer patients, which facilitates precise identification of high-risk individuals and rational formulation of individualized intervention strategies.

The tumor microenvironment (TME) plays a crucial role in ovarian cancer progression and therapy resistance, and ferroptosis has emerged as a key regulator of TME homeostasis. Our single-cell transcriptome analysis revealed distinct ferroptosis activity patterns across different cell components of the ovarian cancer TME, with elevated ferroptosis activity observed in endothelial cells, myeloid cells, fibroblasts, and tumor cells. This indicates that ferroptosis dysregulation is not restricted to malignant tumor cells but also extensively exists in stromal and immune cells, participating in the bidirectional intercellular crosstalk within TME. Stratification based on FRS scores uncovered significant differences in intercellular communication between high-risk and low-risk tumor cells: high-risk tumor cells exhibited more extensive and complex signaling interactions with other TME components through pathways such as ICAM, VEGF, and TIGIT. Such enhanced ligand–receptor crosstalk may drive TME remodeling, thereby facilitating tumor angiogenesis, immune suppression, local invasion and distant metastasis. Immune infiltration analysis further showed that the high-risk group displays markedly elevated immune/stroma scores and higher expression levels of multiple immune checkpoint molecules, indicating an immune-rich rather than immune-desert microenvironment; this phenotype may reflect a dysfunctional, exhausted, or inflammatory immune state, rather than a uniformly immunosuppressive landscape as previously simplified. While the high-risk subgroup harbored increased proportions of activated CD4⁺ memory T cells and regulatory T cells (Tregs) that could promote partial tumor immune escape, the complex immune cell composition precludes reduction to a singular inhibitory phenotype. In contrast, the low-risk group was enriched with M2 macrophages and resting mast cells, which may be associated with a more favorable prognosis. These results underscore the essential regulatory role of ferroptosis in reshaping TME cellular composition and immune landscape, providing novel mechanistic insights into TME-driven ovarian cancer progression. The crosstalk between ferroptosis and the TME also suggests that targeting ferroptosis may not only directly induce tumor cell death but also remodel the immunosuppressive microenvironment, enhancing anti-tumor immune responses.

To elucidate the molecular mechanisms by which FRS regulates ovarian cancer progression, we performed functional enrichment and genomic variation analyses. Gene set enrichment analysis (GSEA) and gene set variation analysis (GSVA) showed that the high-risk subgroup was significantly enriched in tumor-promoting pathways, including NOTCH signaling, TNFA signaling via NF- κ B, and apical junction pathways. These canonical oncogenic pathways are well documented to drive cell proliferation, malignant invasion, inflammatory microenvironment formation and chemoresistance, which mechanistically explains the aggressive phenotype and poor clinical outcome of high-risk patients. In contrast, the low-risk subgroup was enriched in pathways related to MYC targets, oxidative phosphorylation, and DNA repair, reflecting a more stable biological profile associated with better survival outcomes. Survival analysis of pathway activity further confirmed that pathways positively correlated with FRS, such as TNFA/NF- κ B signaling and hypoxia, were associated with poor prognosis, while cholesterol homeostasis pathway, negatively correlated with FRS, was linked to favorable outcomes. Genomic variation analysis revealed distinct mutation and copy number variation (CNV) patterns between FRS subgroups: the high-risk group had higher mutation frequencies of USH2A and NF1 genes, which are involved in tumor progression and RAS signaling regulation, respectively. Additionally, the high-risk group exhibited specific CNV characteristics, including copy number gains in SI, CSMD3, and FLG, and deletions in SYNE2 and NF1. Moreover, combining FRS with intratumoral heterogeneity (MATH score) further improved prognostic stratification, with the high-risk, high-MATH subgroup showing the worst survival outcomes. These findings suggest that genomic mutation, CNV alteration and intratumoral heterogeneity collaboratively modulate FRS prognostic

performance and jointly determine the malignant biological behavior of ovarian cancer. Understanding these molecular mechanisms provides potential therapeutic targets for ovarian cancer, particularly for high-risk patients.

Chemotherapy resistance remains a major clinical challenge in ovarian cancer treatment, and identifying reliable biomarkers for chemotherapy response prediction is critical for personalized treatment.⁴⁵ FRS-defined risk subgroups showed distinct sensitivity to anticancer drugs: low-risk patients were more sensitive to camptothecin, while high-risk patients to A.443654, BMS.708163, Etoposide and PF.4708671, suggesting that FRS can serve as a promising predictive biomarker to guide precise medication selection and avoid ineffective chemotherapy regimens. We verified the synergistic anti-tumor effect of Erastin combined with paclitaxel via in vitro assays in two ovarian cancer cell lines, SKOV3 and Caov-3: Erastin dose-dependently upregulated ferroptosis-related proteins, alleviated paclitaxel resistance, inhibited proliferation and induced apoptosis with BCL2 downregulation and Bax upregulation. Transcriptome analysis revealed DEGs enriched in cell cycle, MAPK and lipid metabolism pathways, confirming the combination exerts anti-tumor effects by regulating ferroptosis, apoptosis and multiple signaling pathways within these two cell models. Crucially, extrapolating this in vitro synergistic effect observed in SKOV3 and Caov-3 cell lines as a universal strategy to alleviate chemotherapy resistance across all ovarian cancer subtypes is premature; the translational scope of this drug combination remains limited, as we only conducted cellular-level in vitro validation without covering ovarian cancer cell lines with diverse pathological subtypes and malignant grades. This preliminary in vitro experimental evidence only supports the exploratory clinical potential of FRS-guided ferroptosis-inducing combined therapy to overcome chemoresistance and improve therapeutic efficacy in ovarian cancer, pending comprehensive preclinical validation including additional cell lines, in vivo animal models and functional rescue assays.

Despite the significant findings of this study, several limitations should be acknowledged. First, the construction and validation of the FRS were based on public database data, including TCGA and GEO datasets. The absence of prospective clinical cohort validation may bring potential selection bias, and the general applicability of FRS in different ethnic populations and real clinical settings still requires further external validation. In our functional assays, ferroptosis was only supported by the expression of ferroptosis-associated proteins; we did not test core markers of lipid peroxidation, namely intracellular iron, MDA and 4-HNE. Furthermore, we did not perform combination index (CI) analysis to quantitatively evaluate the synergy between Erastin and paclitaxel. Second, this study primarily relied on bioinformatics analysis, and the functional verification of core ferroptosis-related genes in the FRS and their specific regulatory mechanisms in ovarian cancer remain unclear. Further in vitro cellular experiments and in vivo animal models are warranted to validate their biological functions in ferroptosis modulation, tumor progression and drug resistance. Third, the study only explored the correlation between FRS and a limited number of chemotherapeutic drugs; the predictive value of FRS for ferroptosis inducers and immune checkpoint inhibitors remains to be systematically explored. Fourth, the single-cell transcriptome analysis was limited to early-stage ovarian cancer patients, and the ferroptosis activity characteristics and TME intercellular communication landscape in advanced and recurrent ovarian cancer remain poorly understood. Finally, the potential role of FRS in other histological subtypes of ovarian cancer, beyond epithelial ovarian cancer, has not been explored. We mainly focused on epithelial ovarian cancer due to insufficient annotated non-epithelial samples for reliable subgroup analysis. Ferroptosis regulation varies drastically across ovarian cancer histological subtypes, and distinct molecular profiles and tumor microenvironments hinder direct extension of FRS to rare subtypes; limited public multi-omics cohorts for non-epithelial tumors also represent a common field-wide constraint.^{46–48}

Conclusion

This study constructs a ferroptosis-related signature (FRS) for ovarian cancer with preliminary prognostic potential, which effectively stratifies patients into high- and low-risk groups with distinct survival outcomes and acts as an independent prognostic biomarker in retrospective public cohorts. The nomogram integrating FRS and clinical features further improves the accuracy of individualized survival prediction. In vitro assays in SKOV3 and Caov-3 ovarian cancer cell lines provide preliminary evidence indicating that Erastin may synergize with paclitaxel to alleviate paclitaxel resistance, possibly via modulating ferroptosis, apoptosis, as well as cell cycle and MAPK signaling pathways. Collectively, the FRS represents a candidate molecular tool for retrospective prognostic stratification and exploratory prediction of chemotherapy response in ovarian cancer, and our correlational multi-omics findings deliver tentative insights into ferroptosis-associated TME remodeling. Notably, the

general clinical applicability of FRS and the translational value of the Erastin-paclitaxel combinatorial regimen remain to be validated in larger independent prospective clinical cohorts and comprehensive preclinical experimental models; further multi-lineage cellular functional rescue assays and in vivo animal verification are required to definitively confirm the mechanistic and therapeutic effects observed herein. Our current work lays a preliminary foundation for follow-up research exploring ferroptosis-targeted strategies to tackle chemotherapy resistance in ovarian cancer.

Data Sharing Statement

These data of this manuscript can be available from the corresponding author Changsong Lin, upon reasonable requests.

Ethics Approval and Consent to Participate

This study did not involve any human subjects, animal experiments, or clinical samples. All transcriptomic and clinical datasets utilized were downloaded from publicly accessible repositories (TCGA, GTEx, GEO), where all participant-related information had undergone strict anonymization and de-identification to eliminate any traceable personal identifiers prior to data submission. Consistent with Item 1 and Item 2 of Article 32 of the national regulatory document Measures for Ethical Review of Life Science and Medical Research Involving Human Subjects (promulgated on February 18, 2023, P.R. China), research solely based on fully anonymized public human omics data without risk to participants' privacy is exempt from institutional ethical review. Accordingly, formal human and animal ethical committee approvals were not applicable to this research.

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Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work. Specifically, Changsong Lin contributed to the conception and design of this study; Minmin Yu provided administrative support; Yidian Zhang was responsible for provision of study materials, data analysis and interpretation; Qianqian Jiang completed collection and assembly of data, data analysis and interpretation; all authors participated in manuscript writing and approved the final submitted manuscript.

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Disclosure

The authors declare no potential conflicts of interest regarding the research, authorship, or publication of this article.

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